Heart Disease Prediction Using Machine Learning

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Problem Statement

Heart disease is the leading cause of death worldwide, causing approximately 17.9 million deaths each year.

Early diagnosis is often costly and time-consuming, which can delay treatment.

Machine learning can help provide fast, low-cost, and accurate predictions.

GOAL: Build a machine learning model to predict heart disease at an early stage.

Dataset Summary

UCI Heart Disease Dataset

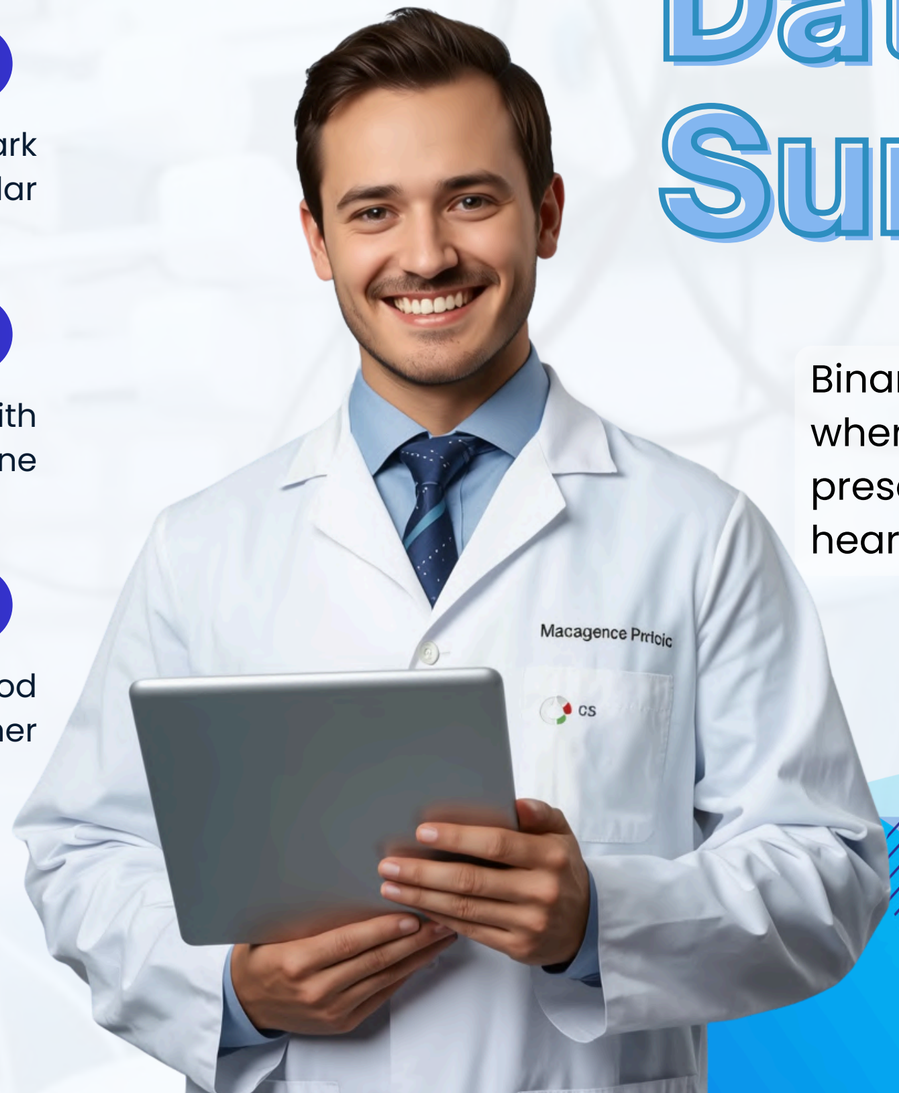
Sourced from Kaggle, this widely-used benchmark dataset contains clinical records for cardiovascular disease prediction and ML research.

303 Patient Records

Dataset includes comprehensive patient data with sufficient samples for training and validating machine learning classification models.

14 Clinical Features

Attributes include age, sex, chest pain type, blood pressure, cholesterol, ECG results, heart rate, and other diagnostic indicators.



Binary classification problem where target variable indicates presence (1) or absence (0) of heart disease.

Preprocessing Steps

Missing Value Handling

Checked dataset for duplicates and missing values and removed 6 incomplete patient records to ensure data quality and model reliability.

Feature Scaling

Applied StandardScaler to normalize all numerical features, ensuring consistent scale across clinical attributes for optimal model performance.

Train-Test Split

Split dataset into 80% training and 20% testing sets for model validation, ensuring unbiased performance evaluation.

EDA Highlights

Key features identified through correlation analysis: cp (chest pain type), thalach (max heart rate), oldpeak (ST depression), ca (major vessels), and thal (thalassemia) show strong predictive power.

Balanced Dataset

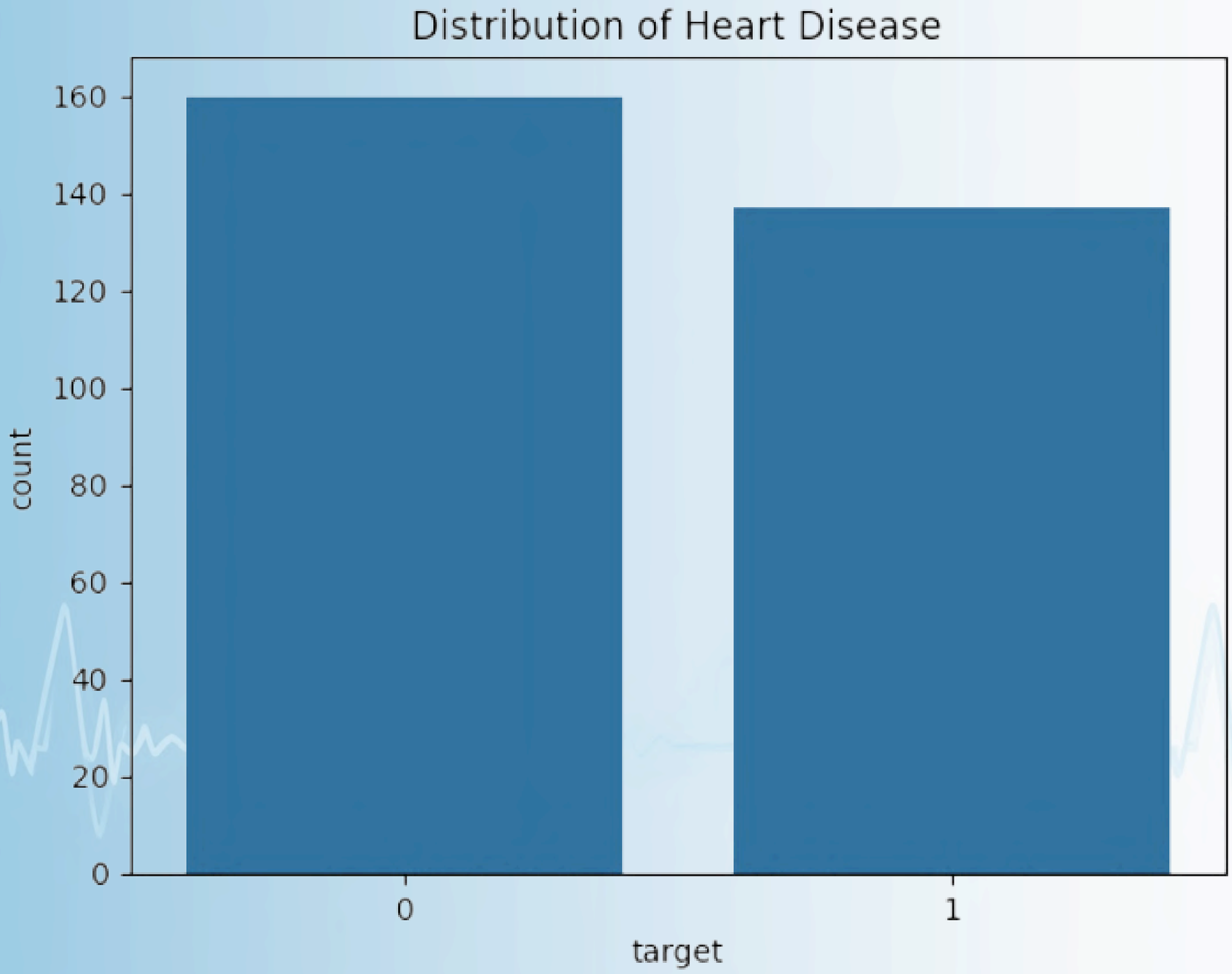
Dataset is well-balanced between disease and no-disease classes, ensuring unbiased model training and reliable predictions.

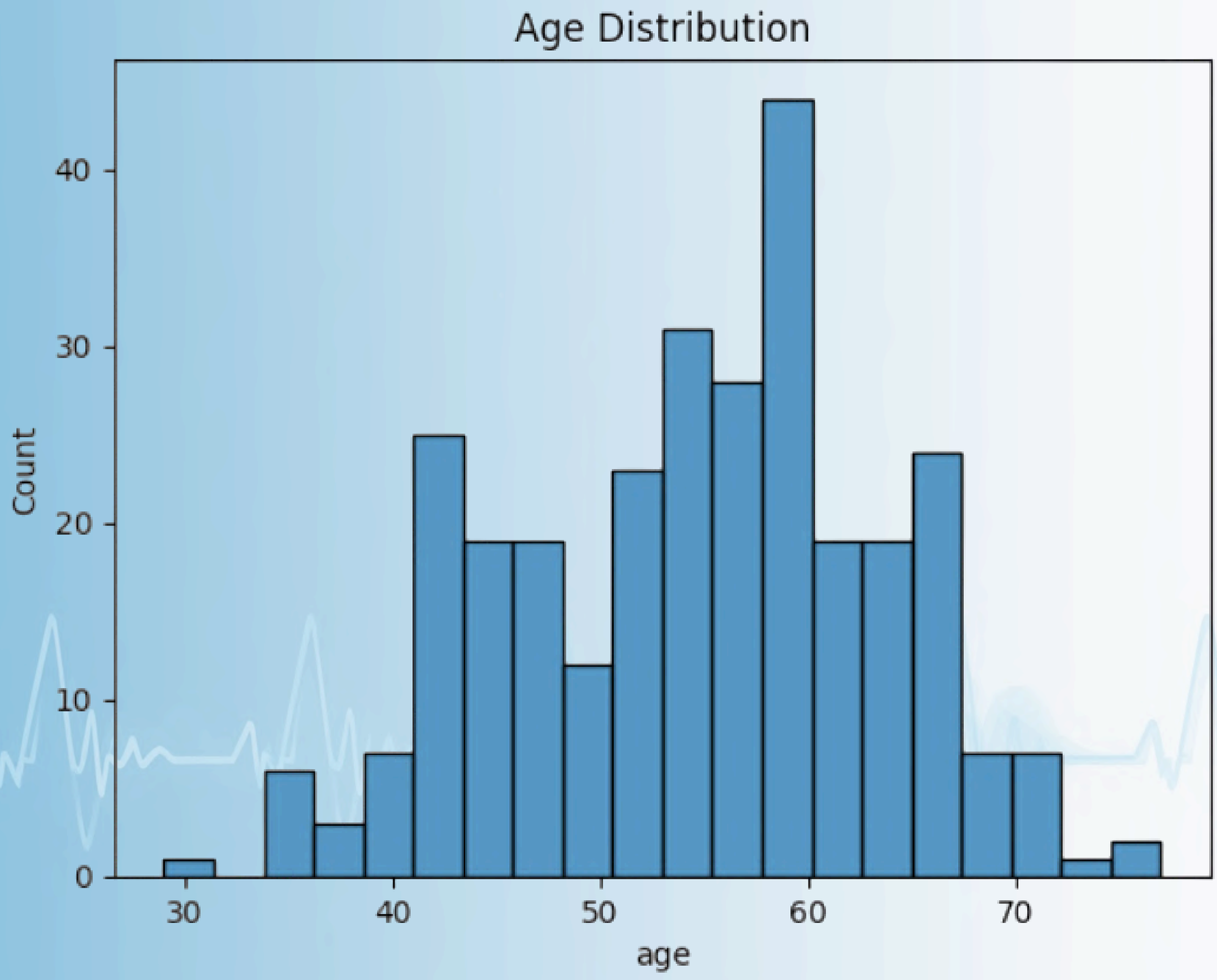
Important Features

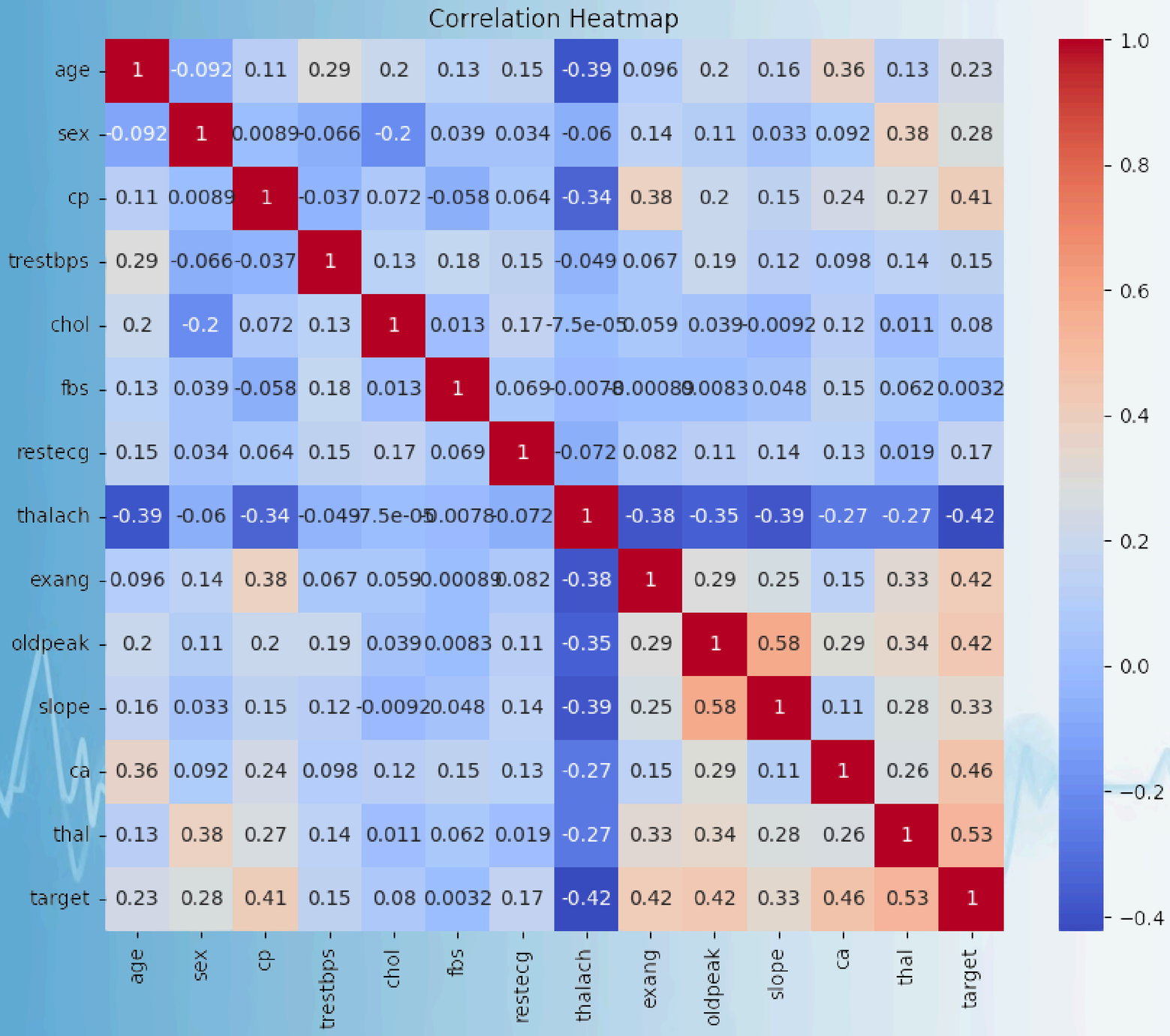
Weak predictors identified: cholesterol (chol) and fasting blood sugar (fbs) showed minimal correlation with heart disease outcome.

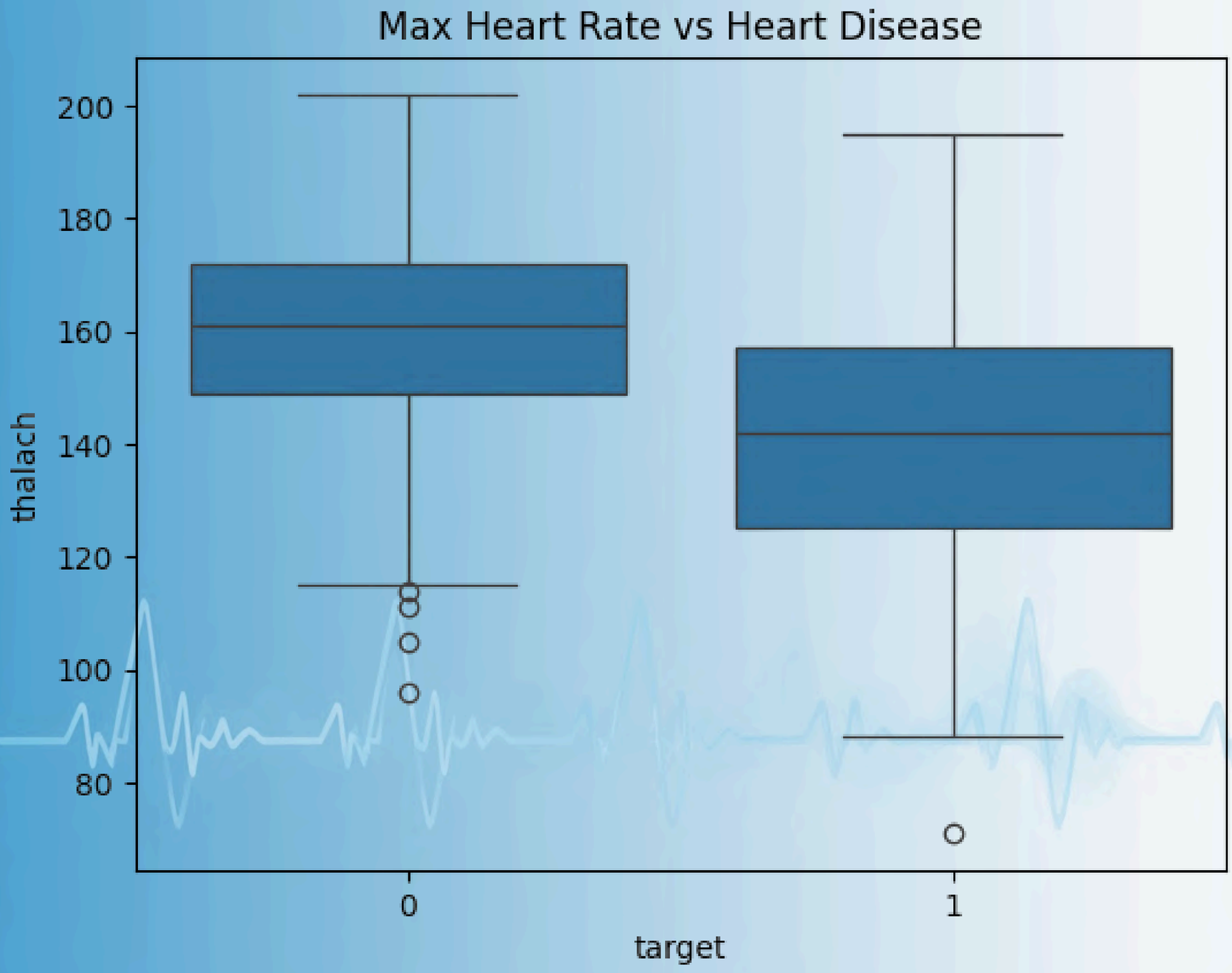
Correlation Analysis

Feature correlation analysis performed to identify relationships between clinical attributes and heart disease presence.


Distribution of Heart Disease
160
140
120
100
80
60
40
20
0
count
0
1
target


Age Distribution
Count
0
10
20
30
40
30
40
50
60
70
age


Correlation Heatmap
age 1 -0.092 0.11 0.29 0.2 0.13 0.15 -0.39 0.096 0.2 0.16 0.36 0.13 0.23
sex -0.092 1 0.0089 -0.066 -0.2 0.039 0.034 -0.06 0.14 0.11 0.033 0.092 0.38 0.28
cp 0.11 0.0089 1 -0.037 0.072 -0.058 0.064 -0.34 0.38 0.2 0.15 0.24 0.27 0.41
trestbps 0.29 -0.066 -0.037 1 0.13 0.18 0.15 -0.049 0.067 0.19 0.12 0.098 0.14 0.15
chol 0.2 -0.2 0.072 0.13 1 0.013 0.17 -7.5e-05 0.059 0.039 -0.0092 0.12 0.011 0.08
fbs 0.13 0.039 -0.058 0.18 0.013 1 0.069 -0.007 -0.00089 0.0083 0.048 0.15 0.062 0.0032
restecg 0.15 0.034 0.064 0.15 0.17 0.069 1 -0.072 0.082 0.11 0.14 0.13 0.019 0.17
thalach -0.39 -0.06 -0.34 -0.049 7.5e-05 -0.007 -0.072 1 -0.38 -0.35 -0.39 -0.27 -0.27 -0.42
exang 0.096 0.14 0.38 0.067 0.059 0.00089 0.082 -0.38 1 0.29 0.25 0.15 0.33 0.42
oldpeak 0.2 0.11 0.2 0.19 0.039 0.0083 0.11 -0.35 0.29 1 0.58 0.29 0.34 0.42
slope 0.16 0.033 0.15 0.12 -0.009 0.048 0.14 -0.39 0.25 0.58 1 0.11 0.28 0.33
ca 0.36 0.092 0.24 0.098 0.12 0.15 0.13 -0.27 0.15 0.29 0.11 1 0.26 0.46
thal 0.13 0.38 0.27 0.14 0.011 0.062 0.019 -0.27 0.33 0.34 0.28 0.26 1 0.53
target 0.23 0.28 0.41 0.15 0.08 0.0032 0.17 -0.42 0.42 0.42 0.33 0.46 0.53 1
age sex cp trestbps chol fbs restecg thalach exang oldpeak slope ca thal target
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4


Max Heart Rate vs Heart Disease
200
180
160
140
120
100
80
thalach
0
1
target

Feature Selection

Strategic feature selection reduced noise and improved model accuracy by focusing on clinically significant predictors.

Selected Key Features

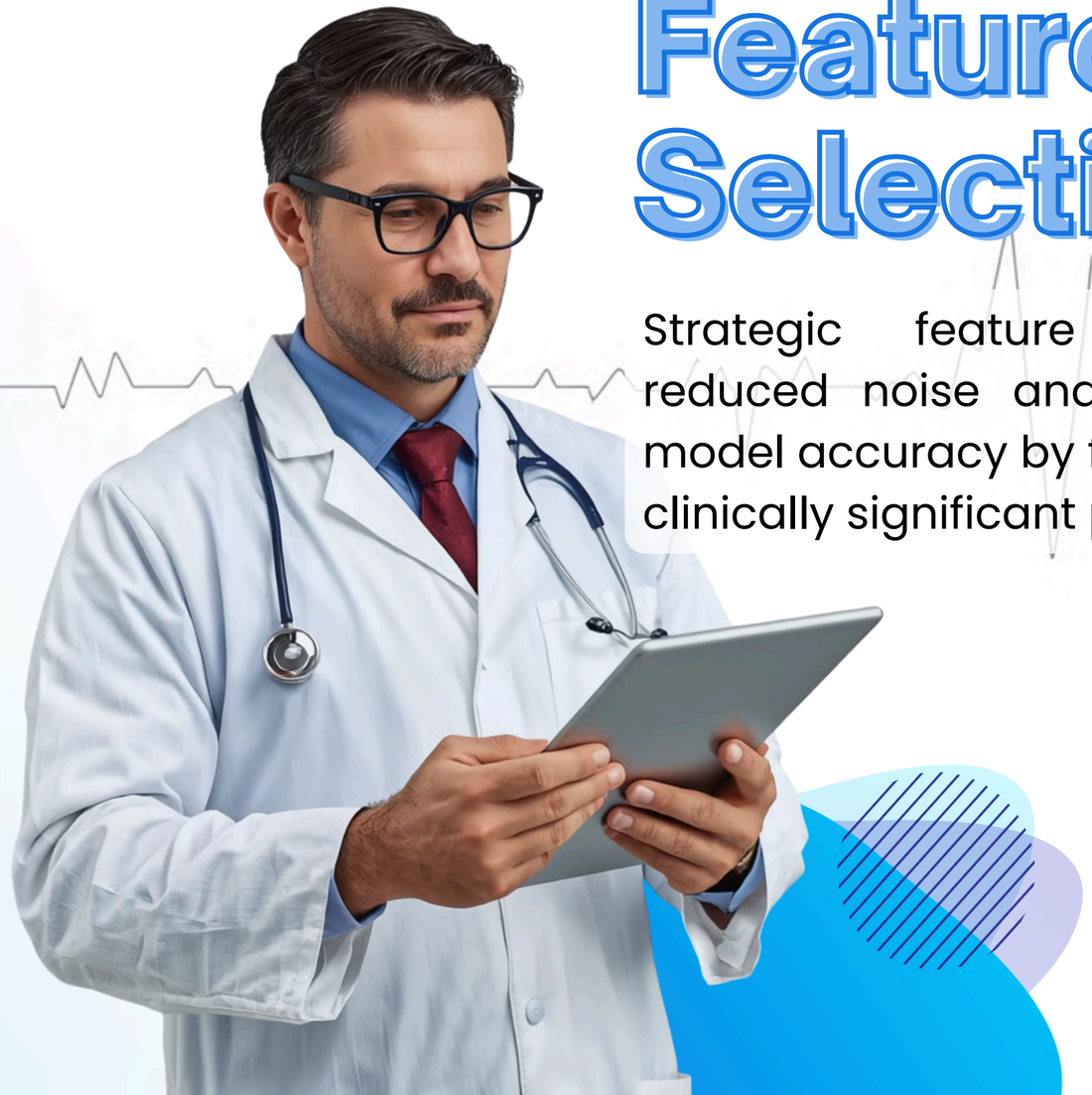
Removed weak features like cholesterol (chol) and fasting blood sugar (fbs) that showed minimal correlation with heart disease outcomes.

Noise Reduction

Applied dimensionality reduction techniques to streamline the model and enhance computational efficiency.

Performance Improvement

Final feature set: thal (thalassemia), ca (major vessels), oldpeak (ST depression), exang (exercise angina), cp (chest pain), thalach (max heart rate).



Models Tested

Decision Tree Classifier

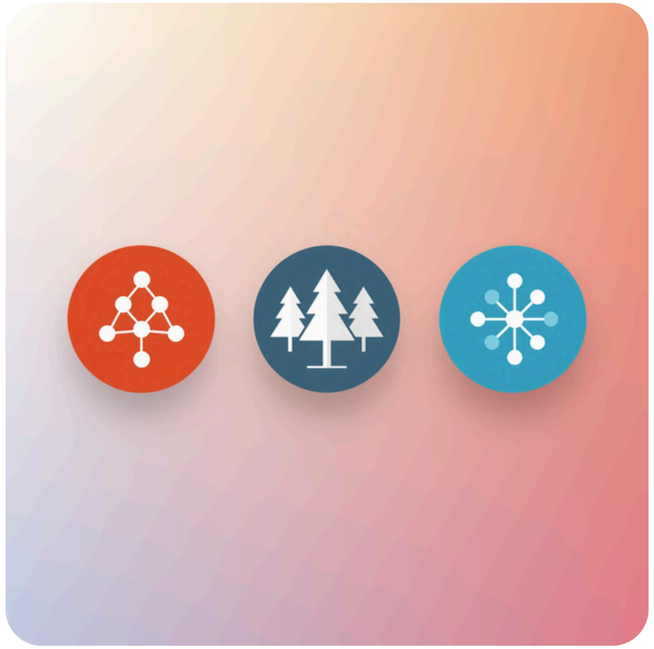
A supervised learning algorithm that splits data based on feature thresholds, creating a tree-like structure for classification decisions.

Random Forest Classifier

An ensemble method combining multiple decision trees to improve accuracy and reduce overfitting through majority voting.

K-Nearest Neighbors (KNN)

A distance-based algorithm that classifies samples based on the majority class of their K nearest neighbors in feature space.



Results Comparison

Three machine learning algorithms were evaluated on the UCI Heart Disease dataset. K-Nearest Neighbors achieved the highest accuracy at 90%, outperforming Random Forest (86.7%) and Decision Tree (76.7%).

KNN's superior performance can be attributed to its effectiveness on small, well-scaled datasets. The feature scaling preprocessing step significantly contributed to these results.

90%

KNN demonstrates strong potential for clinical heart disease screening applications.

Model Evaluation - KNN Performance

K-Nearest Neighbors achieved the highest accuracy among all tested models at 90%. The model demonstrates strong precision (0.95) indicating reliable positive predictions, with good recall (0.79) for sensitivity.

- Accuracy: 90%
- Precision: 0.95
- Recall: 0.79
- F1-Score: 0.86

Final Conclusion

Complete ML Pipeline

Built end-to-end machine learning workflow from data preprocessing to model evaluation. Systematic approach ensures reproducibility and reliability for healthcare applications.

Clinical Application

Model shows strong potential for integration into healthcare screening systems. Early detection capability could support timely medical intervention and improve patient outcomes.

Future Directions

Next steps include testing on larger datasets, exploring deep learning approaches, and validating with real-world clinical data to enhance prediction accuracy and generalizability.

Successfully built a complete ML pipeline for heart disease prediction. KNN achieved best performance with 90% accuracy, demonstrating practical utility for early detection and clinical screening.

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Thank You